Backward Chaining

Exercise 1: Medical Diagnosis

```
% Facts
has_fever(john).
has_cough(john).

% Rule
has_flu(X) :- has_fever(X), has_cough(X).

% Query
% ?- has_flu(john).
```

Output:

```
true.
```

Exercise 2: Student Eligibility Checker

```
% Facts
passed(alice, maths).
passed(alice, science).

% Rule
eligible(X) :- passed(X, maths), passed(X, science).

% Query
% ?- eligible(alice).
```

Output:

```
true.
```

Exercise 3: Travel Planner

```
% Facts
has_ticket(james).
has_visa(james).

% Rule
can_travel(X) :- has_ticket(X), has_visa(X).

% Query
% ?- can_travel(james).
```

Output:

```
true.
```